

Lecture 4

The matrix inversion lemma:

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix}^{-1} = \begin{bmatrix} \bar{E} & F \\ G & H \end{bmatrix}$$

$$\begin{bmatrix} A & B \\ B^T & D \end{bmatrix}^{-1} = \begin{bmatrix} E & F \\ F^T & H \end{bmatrix}$$

$$\begin{cases} \bar{E} = (A - BD^T B^{-1} D)^{-1} \\ F = -\bar{E} B D^{-1} \\ G = -D^{-1} C \bar{E} \\ H = D^{-1} + D^{-1} C \bar{E} B D^{-1} \end{cases}$$

$$E = (A - B D^T B^{-1} D)^{-1}$$

$$F = -\bar{E} B D^{-1}$$

$$H = D^{-1} + D^{-1} B^T \bar{E} B D^{-1} \\ = (D - B^T A^{-1} B)^{-1}$$

This is one of the most important matrix operation for following derivations.

covariance and information:

in lecture 1: $z = h(x) + n$, $n \sim \mathcal{N}(0, R)$ R is noise covariance.

$$\max_x p(z|x) \Leftrightarrow \min_x \|z - h(x)\|_R^2$$

Give the linearisation: $\hat{x}$: $z = h(\hat{x}) + H\tilde{x} + n \Rightarrow \tilde{z} = H\tilde{x} + n$

information $\leftarrow \boxed{H^T R^{-1} H} \tilde{x} = H^T R^{-1} \tilde{z}$

$$\tilde{x} = (H^T R^{-1} H)^{-1} H^T R^{-1} \tilde{z}$$

$$\hat{x}^{\oplus} = \hat{x} + \tilde{x}$$

$$\tilde{z} = H(x - \hat{x}) + n$$

$$= \hat{x} + (H^T R^{-1} H)^{-1} H^T R^{-1} \tilde{z}$$

$$= \hat{x} + (H^T R^{-1} H)^{-1} H^T R^{-1} (H(x - \hat{x}) + n)$$

$$= x + (H^T R^{-1} H)^{-1} H^T R^{-1} n$$

Hence: $P_{x^{\oplus}} = E((x - \hat{x}^{\oplus})(x - \hat{x}^{\oplus})^T)$

$$= (H^T R^{-1} H)^{-1} H^T R^{-1} E(n n^T) R^{-1} H (H^T R^{-1} H)^{-1}$$

$$= (H^T R^{-1} H)^{-1}$$

Hence: Information $\Sigma = H^T R^{-1} H$ $\begin{cases} \Sigma = P^{-1} \\ P = \Sigma^{-1} \end{cases}$
 $P = (H^T R^{-1} H)^{-1}$

This is the fundamental for our following study:

For optimization / least squares, we keep track the information matrix
 For filter / EKF, we keep track of P covariances.

Marginalization:

$x \sim N(\hat{x}, P_x)$: $\hat{x}$ is the best estimate
 P_x is the covariances of x

$x = \begin{bmatrix} x_A \\ x_B \end{bmatrix}$, x_A and x_B are part of the state x .

$x \sim N\left(\begin{bmatrix} \hat{x}_A \\ \hat{x}_B \end{bmatrix}, \begin{bmatrix} P_{AA} & P_{AB} \\ P_{BA} & P_{BB} \end{bmatrix}\right)$ $P^T = P$ P is SPD: symmetric positive definite Matrix
 $P_{AB} = P_{BA}^T$

The distribution of x_A : $x_A \sim N(\hat{x}_A, P_{AA})$
 The distribution of x_B : $x_B \sim N(\hat{x}_B, P_{BB}) \Rightarrow$ This marginalization.

This is marginalization in covariance form: straightforward and easy.

$(H^T R^{-1} H) \tilde{x} = H^T R^{-1} \tilde{z}$
 $\downarrow \quad \downarrow$
 $\Sigma \quad \tilde{x} = b$

let $\tilde{x} = \begin{bmatrix} \tilde{x}_A \\ \tilde{x}_B \end{bmatrix}$, $\begin{cases} \Sigma \text{ is the information matrix} \\ b \text{ is residual} \end{cases}$

How to marginalize $\tilde{x}_A$ or $\tilde{x}_B$?

$\begin{bmatrix} \Sigma_{AA} & \Sigma_{AB} \\ \Sigma_{AB}^T & \Sigma_{BB} \end{bmatrix} \begin{bmatrix} \tilde{x}_A \\ \tilde{x}_B \end{bmatrix} = \begin{bmatrix} b_A \\ b_B \end{bmatrix} \Rightarrow$

how to find $\Sigma'_{AA}, b'_A, \Sigma'_{BB}, b'_B$
 such that: $\begin{cases} \Sigma'_{AA} \tilde{x}_A = b'_A \\ \Sigma'_{BB} \tilde{x}_B = b'_B \end{cases}$

? , , ? , ,

$$i. \Sigma_{AA} = \Sigma_{AA}^T, b_A = b_A$$

let's solve $\Sigma \tilde{x} = b \quad \tilde{x} = \Sigma^{-1} b$

$$\begin{aligned} \tilde{x} = \begin{bmatrix} \tilde{x}_A \\ \tilde{x}_B \end{bmatrix} &= \begin{bmatrix} \Sigma_{AA} & \Sigma_{AB} \\ \Sigma_{AB}^T & \Sigma_{BB} \end{bmatrix}^{-1} \begin{bmatrix} b_A \\ b_B \end{bmatrix} \\ &= \begin{bmatrix} P_{AA} & P_{AB} \\ P_{AB}^T & P_{BB} \end{bmatrix} \begin{bmatrix} b_A \\ b_B \end{bmatrix} \\ &= \begin{bmatrix} P_{AA} b_A + P_{AB} b_B \\ P_{AB}^T b_A + P_{BB} b_B \end{bmatrix} \end{aligned}$$

$$P_{AA} = (\Sigma_{AA} - \Sigma_{AB} \Sigma_{BB}^{-1} \Sigma_{AB}^T)^{-1}$$

$$P_{AB} = -P_{AA} \Sigma_{AB} \Sigma_{BB}^{-1}$$

$$\begin{aligned} P_{BB} &= \Sigma_{BB}^{-1} + \Sigma_{BB}^{-1} \Sigma_{AB}^T P_{AA} \Sigma_{AB} \Sigma_{BB}^{-1} \\ &= (\Sigma_{BB} - \Sigma_{AB}^T \Sigma_{AA}^{-1} \Sigma_{AB})^{-1} \end{aligned}$$

$$\tilde{x}_A = P_{AA} b_A + P_{AB} b_B = P_{AA} b_A - P_{AA} \Sigma_{AB} \Sigma_{BB}^{-1} b_B$$

$$\tilde{x}_A = P_{AA} (b_A - \Sigma_{AB} \Sigma_{BB}^{-1} b_B)$$

$$P_{AA}^{-1} \tilde{x}_A = b_A - \Sigma_{AB} \Sigma_{BB}^{-1} b_B$$

Note: $P_{AB} = -P_{AA} \Sigma_{AB} \Sigma_{BB}^{-1}$

$$= -\Sigma_{AA}^{-1} \Sigma_{AB} P_{BB}$$

$$= -\Sigma_{AA}^{-1} \Sigma_{AB} (\Sigma_{BB}^{-1} + \Sigma_{BB}^{-1} \Sigma_{AB}^T P_{AA} \Sigma_{AB} \Sigma_{BB}^{-1})$$

$$= -(\Sigma_{AA}^{-1} \Sigma_{AB} \Sigma_{BB}^{-1} + \Sigma_{AA}^{-1} \Sigma_{AB} \Sigma_{BB}^{-1} \Sigma_{AB}^T P_{AA} \Sigma_{AB} \Sigma_{BB}^{-1})$$

$$= -(\Sigma_{AA}^{-1} + \Sigma_{AA}^{-1} \Sigma_{AB} \Sigma_{BB}^{-1} \Sigma_{AB}^T P_{AA}) \Sigma_{AB} \Sigma_{BB}^{-1}$$

$$= -(\Sigma_{AA}^{-1} P_{AA}^{-1} + \Sigma_{AA}^{-1} \Sigma_{AB} \Sigma_{BB}^{-1} \Sigma_{AB}^T) P_{AA} \Sigma_{AB} \Sigma_{BB}^{-1}$$

$$= -P_{AA} \Sigma_{AB} \Sigma_{BB}^{-1}$$

$$\tilde{x}_B = P_{AB}^T b_A + P_{BB} b_B = -P_{BB} \Sigma_{AB}^T \Sigma_{AA}^{-1} b_A + P_{BB} b_B = P_{BB} (b_B - \Sigma_{AB}^T \Sigma_{AA}^{-1} b_A)$$

Hence: $P_{BB}^{-1} \tilde{x}_B = b_B - \Sigma_{AB}^T \Sigma_{AA}^{-1} b_A$

$$P_{AA}^{-1} \tilde{x}_A = b_A - \bar{\Sigma}_{AB} \bar{\Sigma}_{BB}^{-1} b_B$$

$$(\bar{\Sigma}_{AA} - \bar{\Sigma}_{AB} \bar{\Sigma}_{BB}^{-1} \bar{\Sigma}_{AB}^T) \tilde{x}_A = b_A - \bar{\Sigma}_{AB} \bar{\Sigma}_{BB}^{-1} b_B$$

$\bar{\Sigma}_{AA}' \tilde{x}_A = b_A' \Rightarrow$ This marginalization for x_B and keep x_A

$$P_{BB}^{-1} \tilde{x}_B = b_B - \bar{\Sigma}_{AB}^T \bar{\Sigma}_{AA}^{-1} b_A:$$

$$(\bar{\Sigma}_{BB} - \bar{\Sigma}_{AB}^T \bar{\Sigma}_{AA}^{-1} \bar{\Sigma}_{AB}) \tilde{x}_B = b_B - \bar{\Sigma}_{AB}^T \bar{\Sigma}_{AA}^{-1} b_A$$

$$\bar{\Sigma}_{BB}' \tilde{x}_B = b_B'$$

This marginalization for x_A and keep x_B

Summary:

For: system

$$\begin{bmatrix} \bar{\Sigma}_{AA} & \bar{\Sigma}_{AB} \\ \bar{\Sigma}_{AB}^T & \bar{\Sigma}_{BB} \end{bmatrix} \begin{bmatrix} \tilde{x}_A \\ \tilde{x}_B \end{bmatrix} = \begin{bmatrix} b_A \\ b_B \end{bmatrix}$$

marginalize x_B and keep x_A : $(\bar{\Sigma}_{AA} - \bar{\Sigma}_{AB} \bar{\Sigma}_{BB}^{-1} \bar{\Sigma}_{AB}^T) \tilde{x}_A = b_A - \bar{\Sigma}_{AB} \bar{\Sigma}_{BB}^{-1} b_B$

Marginalize x_A and keep x_B : $(\bar{\Sigma}_{BB} - \bar{\Sigma}_{AB}^T \bar{\Sigma}_{AA}^{-1} \bar{\Sigma}_{AB}) \tilde{x}_B = b_B - \bar{\Sigma}_{AB}^T \bar{\Sigma}_{AA}^{-1} b_A$

Schur complement

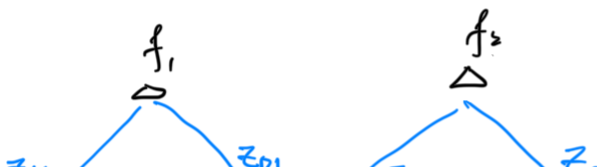
why we care about the information matrix?

Let's use visual measurements for example:

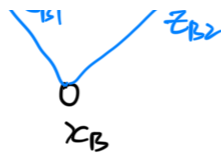
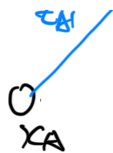
$$z_c = h_c(x) + n$$

$$= h_c(x_c, x_f) + n$$

$$\tilde{z}_c = H_c \tilde{x}_c + H_f x_f + n$$



$$x = \begin{bmatrix} x_A \\ x_B \\ x_C \\ x_f \end{bmatrix}$$



$$\begin{bmatrix} \cdot \\ f_2 \end{bmatrix}$$

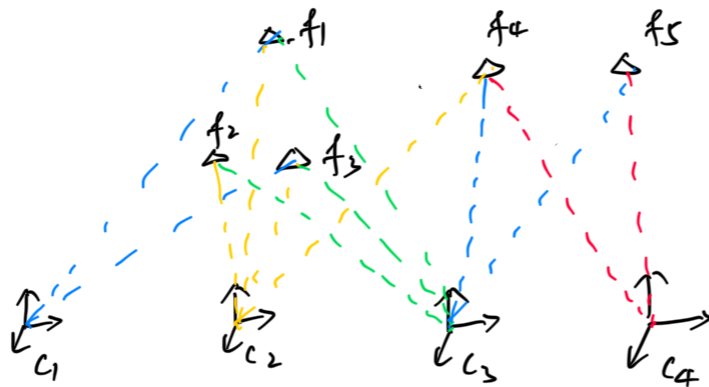
$$\tilde{z} = \begin{bmatrix} \tilde{z}_{A1} \\ \tilde{z}_{B1} \\ \tilde{z}_{B2} \\ \tilde{z}_{C2} \end{bmatrix} = \underbrace{\begin{bmatrix} H_A & 0 & 0 & H_{A1} & 0 \\ 0 & H_B & 0 & H_{B1} & 0 \\ 0 & H_B & 0 & 0 & H_{B2} \\ 0 & 0 & H_C & 0 & H_{C2} \end{bmatrix}}_H \begin{bmatrix} \tilde{x}_A \\ \tilde{x}_B \\ \tilde{x}_C \\ f_1 \\ f_2 \end{bmatrix} + n$$

$n \sim \mathcal{N}(0, I)$

$$\Sigma = H^T H = \begin{matrix} & \begin{matrix} A & B & C & f_1 & f_2 \end{matrix} \\ \begin{matrix} A \\ B \\ C \\ f_1 \\ f_2 \end{matrix} & \begin{bmatrix} \textcircled{X} & 0 & 0 & \textcircled{X} & 0 \\ 0 & \textcircled{X} & 0 & \textcircled{X} & \textcircled{X} \\ 0 & 0 & \textcircled{X} & 0 & \textcircled{X} \\ \textcircled{X} & \textcircled{X} & 0 & \textcircled{X} & 0 \\ 0 & \textcircled{X} & \textcircled{X} & 0 & \textcircled{X} \end{bmatrix} \end{matrix} \cong \begin{bmatrix} \Sigma_{pp} & \Sigma_{pf} \\ \Sigma_{pf}^T & \Sigma_{ff} \end{bmatrix}$$

The information matrix: reflects the problem structure

what's the typical visual odometry / SLAM algorithm?



mono:

① 5 point algorithm:

$$c_1 R, c_1 p_{c2}^*$$

② feature triangulation:

③ P3P, PNP:

$$c_1 R, c_1 p_{c3}$$

④ ...

⑤

5 point algorithm

P3P / PNP

$$c_1 R, c_1 p_{c4}$$

$$f_1, f_1, f_2 \dots$$

BA with:

$$c_1, c_1, c_2$$

$$f_1, f_1, f_2 \dots$$

BA with:

$$c_1, c_2, c_3, c_4$$

$$f_1, f_1, f_2$$

⑦ when we have too many poses $c_1, \dots, c_n$ we can marginalise $c_1, f_1, \dots$
 $f_1, \dots, f_m$
 to make the system light-weight.

Now: we have built up the visual SLAM with last two courses.

Discussion: $z = h(x) + n$ $n \sim \mathcal{N}(0, R)$

① Typical solution: $\min_x \|z - h(x)\|_R^2$

GN or LM: Given linearization $\hat{x}$, $\tilde{z} = H\tilde{x} + n$

$\hat{x}^{(1)} = \hat{x}^0 + \tilde{x}$, then redo linearization $\tilde{z} = H^0\tilde{x} + n$

② what if we just linearize once:

$$\hat{x}^0, \quad z = h(\hat{x}^0) + H^0(x - \hat{x}^0) + n \quad H^0 = H|_{\hat{x}^0}$$

$$z - h(\hat{x}^0) = H^0(x - \hat{x}^0) + n$$

$$= H^0(x - \hat{x} + \underbrace{\hat{x} - \hat{x}^0}_{\Delta x}) + n$$

$$= H^0\tilde{x} + H^0\Delta x + n$$

$$z - h(\hat{x}^0) - H^0\Delta x = H^0\tilde{x} + n$$

$$\tilde{z} - \boxed{H^0\Delta x} = H^0\tilde{x} + n$$

residual correction.

△ FEJ: first-estimate Jacobians, OKVIS

③ For $x = \begin{bmatrix} x_A \\ x_B \end{bmatrix}$, $z = h(x_A, x_B) + n$

$$z = h(\hat{x}_A, \hat{x}_B^0) + H_A \tilde{x}_A + H_B^0 \tilde{x}_B + \underbrace{H_B^0 \Delta x_B + n}$$

Hence: $\underline{z} = h(\hat{x}_A, \hat{x}_B^0) + H_B^0 \Delta x_B = H_A \tilde{x}_A + H_B^0 \tilde{x}_B + n$

$$\tilde{z}^0 - H_B^0 \Delta x_B = [H_A \ H_B^0] \begin{bmatrix} \tilde{x}_A \\ \tilde{x}_B \end{bmatrix} + n$$

△: pre-integration, ACI: analytic combined integration

iSAM: also uses similar idea.

④ For $x = \begin{bmatrix} x_A \\ x_B \end{bmatrix}$ $z = h(x_A, x_B) + n$
 $\tilde{z} = H_A \tilde{x}_A + H_B \tilde{x}_B + n$

If we can find N that $N^T H_B = 0$, then:

$$N^T \tilde{z} = N^T H_A \tilde{x}_A + \underbrace{N^T H_B \tilde{x}_B}_0 + N^T n$$

Algorithm: MSCKF, Smart vision factor / structure-less vision factor.

⑤ For $x = \hat{x} + \tilde{x}$, but if we put x into $SE(3)$, $SE_2(3)$ and only right errors:

$$x = \hat{x} \boxplus \tilde{x} = \underbrace{\text{Exp}(\tilde{x})}_{\text{right}} \cdot \hat{x} \rightarrow \text{invariant SLAM/VIO}$$

⑥: For: $x = \begin{bmatrix} x_A \\ x_B \end{bmatrix}$, we define: $\left\{ \begin{array}{l} x_A = \hat{x}_A \boxplus \tilde{x}_A = \text{Exp}(\tilde{x}_A) \cdot \hat{x}_A \\ x_B = \hat{x}_B + \tilde{x}_B \end{array} \right\}$

Decoupled right invariant SLAM/VIO

Use your imagination to deal with $\left\{ \begin{array}{l} z = h(x) + n \\ \tilde{z} = H \tilde{x} + n \end{array} \right\}$ to design different algorithms

How to design a common system:

$$p(x, \hat{x}, P)$$

→ prior

$$\begin{cases} x_0 \sim N(\mu_0, \Sigma_0) \\ x_i = f(x_0) + w_0 \quad w_k \sim N(0, Q) \rightarrow \text{motion model} \\ z_i = h(x_i) + r_i \quad r_i \sim N(0, R) \rightarrow \text{measurement model} \end{cases}$$

$$\max_x p(x|z) = \max_x \frac{p(z|x)p(x)}{p(z)} = \max_x \frac{p(z|x)p(x|x_0)p(x_0)}{p(z)}$$

$$\Leftrightarrow \min_x p(z|x)p(x|x_0)p(x_0)$$

$$\Leftrightarrow \min_x \|z_i - h(x_i)\|_R^2 + \|x_i - f(x_0)\|_Q^2 + \|x_0 - \hat{x}_0\|_{P_0}^2$$

Two way solve it:

① batch solving: $\|x_0 - \hat{x}_0\|_{P_0}^2 + \|x_i - f(x_0)\|_Q^2 + \|z_i - h(x_i)\|_R^2 \Rightarrow \hat{x}_i, P_i$

② incremental solving: ① $\|x_0 - \hat{x}_0\|_{P_0}^2 + \|x_i - f(x_0)\|_Q^2$

② marginalise x_0 , keep $x_i, (\hat{x}_{i|0}, P_{i|0})$

③ $\|x_i - \hat{x}_i\|_{P_{i|0}}^2 + \|z_i - h(x_i)\|_R^2 \Rightarrow \hat{x}_i, P_i$

② Solve the incremental way:

$$x_i = f(x_0) + w_0 \Rightarrow \hat{x}_i + \tilde{x}_i = f(\hat{x}_0) + F\tilde{x}_0 + w_0 \Rightarrow r_x \triangleq \hat{x}_i - f(\hat{x}_0) = F\tilde{x}_0 - \tilde{x}_i + w_0$$

$$\min_x \|\tilde{x}_0\|_{P_0}^2 + \|r_x - [F - I_2] \begin{bmatrix} \tilde{x}_0 \\ \tilde{x}_i \end{bmatrix}\|_{Q_0}^2$$

$$= \tilde{x}_0^T P_0^{-1} \tilde{x}_0 + \begin{bmatrix} \tilde{x}_0 \\ \tilde{x}_i \end{bmatrix}^T \begin{bmatrix} F^T Q_0^{-1} F & -F^T Q_0^{-1} \\ -Q_0^{-1} F & Q_0^{-1} \end{bmatrix} \begin{bmatrix} \tilde{x}_0 \\ \tilde{x}_i \end{bmatrix} - 2 \begin{bmatrix} \tilde{x}_0 \\ \tilde{x}_i \end{bmatrix}^T [F - I_2]^T Q_0^{-1} r_x + r_x^T Q_0^{-1} r_x$$

$$C = \begin{bmatrix} \tilde{x}_0 \\ \tilde{x}_i \end{bmatrix}^T \begin{bmatrix} P_0^{-1} + F^T Q_0^{-1} F & -F^T Q_0^{-1} \\ -Q_0^{-1} F & Q_0^{-1} \end{bmatrix} \begin{bmatrix} \tilde{x}_0 \\ \tilde{x}_i \end{bmatrix} - 2 \begin{bmatrix} \tilde{x}_0 \\ \tilde{x}_i \end{bmatrix}^T [F - I_2]^T Q_0^{-1} r_x + r_x^T Q_0^{-1} r_x$$

$$\Leftrightarrow \frac{\partial C}{\partial \begin{bmatrix} \tilde{x}_0 \\ \tilde{x}_i \end{bmatrix}} = 0 \Rightarrow \begin{bmatrix} P_0^{-1} + F^T Q_0^{-1} F & -F^T Q_0^{-1} \\ -Q_0^{-1} F & Q_0^{-1} \end{bmatrix} \begin{bmatrix} \tilde{x}_0 \\ \tilde{x}_i \end{bmatrix} = \begin{bmatrix} F^T \\ -I_2 \end{bmatrix} Q_0^{-1} r_x$$

$$\begin{aligned}
&= (P_{110} - P_{110} H^T (H P_{110} H^T + R)^{-1} H P_{110}) \tilde{z} \\
&= (P_{110} H^T R^{-1} - P_{110} H^T (H P_{110} H^T + R)^{-1} H^T P_{110} H^T R^{-1}) \tilde{z} \\
&= P_{110} H^T (R^{-1} - (H P_{110} H^T + R)^{-1} H^T P_{110} H^T R^{-1}) \tilde{z} \\
&= P_{110} H^T (H P_{110} H^T + R)^{-1} ((H P_{110} H^T + R) R^{-1} - H^T P_{110} H^T R^{-1}) \tilde{z} \\
&= P_{110} H^T (H P_{110} H^T + R)^{-1} \tilde{z}
\end{aligned}$$

Finally:
$$\begin{cases} \tilde{x}_t = P_{110} H^T (H P_{110} H^T + R)^{-1} \tilde{z} \\ P_{111} = P_{110} - P_{110} H^T (H P_{110} H^T + R)^{-1} H P_{110} \end{cases}$$

The relationship between schur complement and null space operation:

$$z = h(x_A, x_B) + n \quad n \in N(0, R)$$

$$\tilde{z} = H_A \tilde{x}_A + H_B \tilde{x}_B + n$$

① information form:

$$\begin{bmatrix} H_A^T R^{-1} H_A & H_A^T R^{-1} H_B \\ H_B^T R^{-1} H_A & H_B^T R^{-1} H_B \end{bmatrix} \begin{bmatrix} \tilde{x}_A \\ \tilde{x}_B \end{bmatrix} = \begin{bmatrix} H_A^T \\ H_B^T \end{bmatrix} \tilde{z}$$

marginalize $\tilde{x}_B$, keep $\tilde{x}_A$:

Eq ①

$$\underbrace{(H_A^T R^{-1} H_A - H_A^T R^{-1} H_B (H_B^T R^{-1} H_B)^{-1} H_B^T R^{-1} H_A)}_{\text{Information}} \tilde{x}_A = \underbrace{H_A^T \tilde{z} - H_A^T R^{-1} H_B (H_B^T R^{-1} H_B)^{-1} H_B^T \tilde{z}}_{\text{residual}}$$

U: upper triangular matrix

② null space form:

$$\tilde{z} = H_A \tilde{x}_A + H_B \tilde{x}_B + n \quad \text{if} \quad H_B = \begin{bmatrix} 0_e & 0_n \end{bmatrix} \begin{bmatrix} U \\ 0 \end{bmatrix}$$

$$= 0_e U + 0_n \cdot 0 = 0_e U$$

$$m^T \tilde{z} = 0_m^T H_A \tilde{x}_A + 0_n^T H_B \tilde{x}_B + 0_n^T n$$

$$n \cdot 11 = n$$

$$y = Hx + n$$

$$= \begin{bmatrix} 0 & 1 \end{bmatrix} H_A \tilde{x}_A + 0 + \begin{bmatrix} 0 & 1 \end{bmatrix} n$$

$$y = Hx + n$$

$$\begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 \end{bmatrix}^T = 0 \cdot 0 + 1 \cdot 1 = 1$$

$$\begin{bmatrix} 0 & 1 \end{bmatrix} H_A \tilde{x}_A + 0 + \begin{bmatrix} 0 & 1 \end{bmatrix} n = \begin{bmatrix} 0 & 1 \end{bmatrix} H_A \tilde{x}_A + \begin{bmatrix} 0 & 1 \end{bmatrix} n$$

$$\text{where } \tilde{x}_A \triangleq \tilde{x}_B$$

$$\text{If } R = \sigma^2 I: \text{ with } H_B = 0 \cdot U, \text{ then:}$$

$$H_B^T R^{-1} H_B = (0 \cdot U)^T \frac{1}{\sigma^2} \cdot 0 \cdot U = \frac{U^T U}{\sigma^2}$$

$$\text{Then: } H_A^T R^{-1} H_B (H_B^T R^{-1} H_B)^{-1} H_B^T R^{-1} H_A$$

$$= H_A^T \frac{1}{\sigma^2} \cdot 0 \cdot U \cdot \frac{U^T U}{\sigma}$$